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Exploring the nexus of occupational licensing and the shadow economy: Evidence from U.S. states

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Abstract

This study looks at the relationship between the prevalence of state-level shadow economies and the extent of labor regulations specific to occupational licensing. Occupational licensing leads to higher prices in the formal sector and barriers to entry into formal sector employment, therefore incentivizing individuals to migrate their demand and supply to the underground sector. Using state-level panel data for the 50 U.S. states from 2001 to 2019 and two-stage least squares estimation, we find a positive relationship between occupational licensing in a state and the size of its shadow economy. These results are robust to various considerations.

KEYWORDS
occupational licensing, shadow economy, U.S. states

JEL CLASSIFICATION
J44, K42, K31, L51, O51

1 | INTRODUCTION

Occupational licensing in the U.S. has become much more prevalent over the last several decades. In the 1950's approximately 5% of occupations required a license to practice in the U.S. and that estimate has ballooned in 2022 to 22%.[1] An occupational license is defined by Kleiner and Vorotnikov (2017: pg. 133–134) as "the legal process by which governments (mostly U.S. states but also local governments and the federal government) identify the legal qualifications required to work in a trade or profession, after which only regulated practitioners are allowed by law to receive pay for doing tasks in the occupation." Occupations that are more dominated by licensing requirements include healthcare practitioners (72.6% of workforce), legal (61.3% of workforce), and education (51%).[2]

While occupational licensing affects roughly 20% of U.S. workers there is considerable variation across U.S. states (Trudeau & Timmons, 2023). Trudeau and Timmons (2023) discuss the specific example of a "shampooer," which is a profession that requires a cosmetology license in some states, no licenses in other states, and a specific shampooer license in yet other states. Based on a comprehensive measure of occupational licensing across U.S. states (and D.C.) from the Archbridge Institute, Arkansas, Texas, Alabama, Oklahoma, and Washington are the least restrictive states, whereas Colorado, Indiana, Wyoming, Missouri, and Kansas are the most restrictive states.[3] While occupational licensing requirements seem obvious for some professions, for example, a physician or attorney, other professions are less obvious. For example, Harrigan (2023) discusses that in Kansas, the hair-removal process known as "sugaring," which uses a nonhazardous mixture of sugar, water, and lemon juice to remove hair requires an esthetician license (or

Abbreviations: DGE, dynamic general equilibrium; MIMIC, multiple indicators, multiple causes; UKCPR, UK Center for Poverty Research.

cosmetology license) which involves as much as 1000 hours of education.[4] Many other examples of lower-income occupations that require a license in nearly all states include a cosmetologist, bus driver, manicurist, pest control applicator, milk sampler, makeup artist, animal breeder, auctioneer, etc. (Knepper et al., 2022). Among lower-income occupations that require a license the cost averages $295 in fees, at least one exam, and almost a year's worth of education and experience (Knepper et al., 2022).[5]

As with any labor market regulation, occupational licensing comes with both costs and benefits. The primary justification for occupational licensing is to mitigate problems with asymmetric information to ensure the quality of the service and protect the health and safety of the consumer (Kleiner & Vorotnikov, 2017). Licensing, however, restricts employment opportunities by raising barriers to entry into licensed professions thereby raising consumer costs through reduced competition (Knepper et al., 2022). For example, Kleiner (2015) states that occupational licensing may cost the U.S. economy as much as 2.85 million jobs and costs consumers as much as $203 billion.

Another potential consequence of occupational licensing that is often overlooked in the literature is the extent to which these barriers to entry encourage workers and consumers to avoid the formal sector and engage in the underground, or shadow, economy—see Samuel et al. (2021) for an important exception.[6] According to Samuel et al. (2021: pg. 325) "licensing creates scope for an 'underground economy'of firms that operate without a license, thereby allowing them to bypass quality standards entirely." Labor regulations, including occupational licensing, contribute to labor costs and have been identified as one of the main determinants of the shadow economy across countries (see, e.g., Johnson et al., 1998; Friedman et al., 2000) and across U.S. states (Wiseman, 2013). Licensing requirements, especially among lower-income occupations such as barbers, landscapers, and dog breeders, are particularly burdensome for individuals on the margins of the formal and informal sectors.

Evidence suggests that informal employment is a result of being excluded from the formal sector and individuals voluntarily exiting the formal sector (see OECD, 2009; Schneider & Enste, 2013). On the demand side, labor restrictions lead to higher prices for a particular good or service and, in turn, encourage consumers to seek out alternatives in underground markets where it has been shown that prices are relatively lower and the quality is comparable (Schneider & Enste, 2013). Empirically, Wiseman (2013) shows that U.S. states with less restrictive labor regulations (more labor market freedom) are associated with smaller shadow economies.

Given the obvious link between occupational licensing requirements and the size of the shadow economy, a natural question arises: To what extent does occupational licensing contribute to the size of the shadow economy? In answering this question, one problem identified by Samuel et al. (2021) is the lack of good measures of the shadow economy due to its inherent clandestine nature. Therefore, to answer this question and overcome this barrier, we use a novel measure of the U.S. state-level shadow economy from Marshall et al. (2023). Marshall et al. (2023) use a two-sector dynamic general equilibrium (DGE) model to estimate the size of the state-level shadow economy (% of GDP) for the 50 U.S. states from 1999 to 2019. This measure of the shadow economy improves on prior estimates of state-level shadow economies from Wiseman (2013), given its theoretical footing and more recent time coverage. Based on this measure of the shadow economy, the average size of the state-level shadow economy is 12% of GDP and, the same with occupational licensing, varies considerably across states. Using this measure of the shadow economy, we estimate the relationship between occupational licensing and the size of the shadow economy for the 50 U.S. states from 2001 to 2019. The results, based on an instrumental variables technique and two-stage least squares estimation, show that, on average, states with a higher prevalence of occupational licensing have larger shadow economies.

The paper proceeds as follows: Section 2 reviews the literature and derives a testable research hypothesis; Section 3 documents the data, model, and estimation; Section 4 discusses the results; and Section 5 concludes.

2 | LITERATURE REVIEW AND HYPOTHESES

This paper blends two literatures together including the effects of labor restrictions associated with occupational licensing and the determinants of the shadow economy. For general overviews of occupational licensing see Rottenberg (1962), Kleiner (2000), Kleiner and Krueger (2013), and Larkin (2016), and for the shadow economy see Schneider and Enste (2000, 2013), Gërxhani (2004), Schneider (2011), Goel et al. (2019), and Berdiev and Saunoris (2020).

Related to Becker's (1968) work on the economics of crime, individuals undergo a cost-benefit analysis when deciding whether to engage in the shadow economy. This cost-benefit calculus is primarily influenced by taxes, regulations, and institutional quality (see Schneider & Enste, 2000, 2013). In this paper, we argue that occupational licensing influences individuals' and businesses' decision to operate in the shadow economy by providing services illegally without a license (the supply side) as well as consumers' decision to seek out these unregulated goods and services (the demand side).

The main drawback of occupational licensing is that it restricts the labor supply by creating a barrier to entry into specific fields or occupations (see, e.g., Federman et al., 2006; Hall et al., 2018). Specifically, Blair and Chung (2019) find that a 17%–27% reduction in the labor supply is due to occupational licensing. Prospective workers or entrepreneurs who are unable to obtain a license are then forced to abandon their chosen profession or operate in the underground economy illegally without a license (Chambers et al., 2019). If the benefit (i.e., cost savings from forgoing the license) of working underground without a license outweighs the direct cost (i.e., fine/jail) and indirect, or opportunity, cost (i.e., formal employment), then the individual will choose to migrate to the shadow economy. Indeed, labor regulations have been shown to be one important determinant of the shadow economy (Blanton & Peksen, 2019). Occupational licensing is one specific type of labor regulation that may impact the size of the shadow economy either by forcing individuals into the shadow economy or preventing shadow workers from transitioning to the formal sector (see Wiseman, 2013; Wiseman & Walker, 2017). In fact, states with high labor market barriers tend to be associated with larger shadow economies (Wiseman, 2020). In developing one of the first estimates of the size of the shadow economy at the state level for the 50 U.S. states from 1997 to 2008 using the Multiple Indicators, Multiple Causes (MIMIC) method, Wiseman (2013) finds that the main determinants of the size of the shadow economy include social welfare burdens, labor market regulations and the intensity of enforcement.

Using data from CPS and SIPP, Kleiner and Xu (2020) find that occupational licensing serves as a barrier to entry disproportionally affecting employed workers more than unemployed workers due to the higher opportunity cost associated with employed workers. Arguably, the same line of reasoning applies to workers employed informally such that the higher opportunity cost reduces the ability of informal workers to transition to formal employment. In fact, the empirical evidence supports a high labor substitution between formal and informal sectors in the economy (Lemieux et al., 1994). This is especially problematic given that many licenses are required for entry-level positions (de Rugy, 2014), which disproportionally impacts younger and lower-income individuals who are more susceptible to underground work. Not surprisingly, many lower-income occupations that often require a license, such as barber, cosmetologist, landscaper, home childcare provider, carpenter, animal breeder, taxidermist, dog trainer, tree trimmer, to name a few, are also more prone to underground work.[7] Based on opinion polls from Austria and Germany, some of the more popular occupations for illicit work include things like hairdressing, child care, car repairs, housework, and building—see Figures 2.3 and 2.4 in Schneider and Enste (2013). Using data from U.S. states, Hall and Pokharel (2016) find an inverse relationship between occupational licensing for barbers and the number of barber shops, which may suggest that at least some of these barbers chose to forgo the license by moving underground. Furthermore, individuals with a criminal record are either banned from obtaining a license or face other obstacles in obtaining a license and therefore may be forced into the underground economy to make ends meet (Carpenter et al., 2017).

The main rationale for occupational licensing is to reduce the asymmetric information in the labor market by confirming an individual's qualifications in a specific occupation, thereby improving the quality of the good or service (Kleiner & Soltas, 2023; Samuel et al., 2021).[8] Higher quality goods/services in the formal sector, relative to the informal sector, reduces the incentive (i.e., raises the opportunity cost) to demand the same goods or services in underground markets. However, there is a lack of empirical evidence supporting the link between occupational licensing and quality improvements (Kleiner, 2000) and, in some cases, show a decrease in quality (Carroll & Gaston, 1981). Carpenter (2012), for example, finds a lack of statistical difference in how licensed and non-licensed florists rate floral arrangements. Similarly, Kleiner and Kudrle (2000) find that among Air Force recruits there was a lack of statistical support for more restrictive licensing requirements in the dental field improving dental health. Finally, Farronato et al. (2020) find that while more restrictive occupational licensing contributes to reduced competition and higher prices, consumer satisfaction is not improved. The lack of evidence tying occupational licensing to improvements in quality suggests that the opportunity costs of demanding underground services are low.

As a consequence of labor restrictions related to licensing and the reduction in the labor supply, wages of licensed workers tend to be higher, which in turn drives prices higher (Kleiner, 2006; Kleiner et al., 2016; Kleiner & Vorotnikov, 2017; Pagliero, 2010). Gittleman and Kleiner (2016) use data from the National Longitudinal Survey of Youth to find that workers with an occupational license have higher pay relative to unlicensed workers in the U.S. Some estimates show that wages are as much as 19% higher for licensed workers (Kleiner & Krueger, 2010, 2013). More recently, Kleiner and Soltas (2023) find that going from a licensed state to an unlicensed state, a licensed occupation experienced a 15% increase in the average wage as well as a 3% increase in number of hours per worker, while also reducing employment by 29%.

Higher prices will increase the incentive for consumers to seek out underground alternatives where prices are estimated to be a third to a quarter less than the formal sector price and of similar quality (Schneider & Enste, 2013). In fact, one of the main reasons for the demand of goods and services produced underground, as identified by survey data from Germany, is the higher prices in the formal sector (see Table 6.2 in Schneider & Enste, 2013). In summary, high prices and negligible differences in quality serve as an incentive for individuals to opt for goods and services from the underground economy over those from the formal economy.

Other side effects of labor restrictions include its influence on income inequality and inter-state migration. Chambers et al. (2019) find evidence, based on a panel of 115 countries, that more stringent labor market entry regulations, including licensing, are linked to greater income inequality, while greater income inequality has been shown to be associated with larger shadow economies (Berdiev & Saunoris, 2019). It has also been shown that state-specific occupational licensing reduces worker inter-state mobility (Johnson & Kleiner, 2020) and these added restrictions on mobility would incentivize individuals to bypass inter-state restrictions by working illegally. For example, an individual licensed in a given state may relocate to a state without reciprocity, where their license is invalid, potentially forcing them to work underground.

To date, scant research has been done on the connection between occupational licensing and the shadow economy, in part because of the lack of good measures of the shadow economy (Samuel et al., 2021). One important exception is that of Samuel et al. (2021) who show theoretically that licensing has an ambiguous effect on underground housing and is dependent on the chosen parameters; however, they find that strengthening enforcement tends to reduce underground housing activities and encourage landlords to increase the quality of their properties. In this paper, we take a more macro approach to fill this gap in the literature, by using a relatively new measure of the state-level shadow economy from Marshall et al. (2023) to ascertain the empirical relationship between the shadow economy and occupational licensing at the state level.

Based on the aforementioned theoretical and empirical evidence we hypothesize the following:

Hypothesis. Occupational licensing is associated with larger shadow economies, ceteris paribus.

3 | EMPIRICAL MODEL, DATA, AND ESTIMATION

3.1 | Empirical model

To test our research hypothesis, we use the following linear model:

$$\text{Shadow Economy}_{it} = \beta_0 + \beta_1 \text{Occupational Licensing}_{it} + \gamma X_{it} + \mu_i + \tau_t + \varepsilon_{it} \tag{1}$$

where i denotes a U.S. state and t denotes a year. The dependent variable is the size of the shadow economy as a percentage of GDP; the main independent variable of interest is Occupational Licensing; μ_i is the state-specific fixed effects; τ_t captures time effects; and ε_{it} is the random error term.

The vector X includes the control variables. First, more prosperous states, measured by the log of GDP per capita (LnGDP), means more opportunities for formal sector employment, which reduces the incentive to produce in the shadow economy (Goel & Nelson, 2016; Marshall et al., 2023). Secondly, larger state and local governments (Government Size), measured by total state and local government spending as a percentage of GDP, may signify more burdensome governments, with factors such as higher taxes incentivizing individuals to engage in underground activities. On the other hand, governments with greater resources are more powerful in their ability to combat activities in the shadow economy (Goel & Nelson, 2016; Goel et al., 2019; Marshall et al., 2023). Third, a more educated populace, measured by the percentage of the population with a bachelor's degree or more (Education), raises the opportunity cost of working underground (Berdiev et al., 2015). Finally, a larger union presence in a state, measured by the percentage of workers that are covered by a collective bargaining agreement (Union Coverage) restricts the labor supply forcing individuals to move underground (Berdiev et al., 2015).

3.2 | Data

This study uses state-level panel data for the 50 U.S. states from 2001 to 2019—see Table 1 for the variable names, descriptions, and sources and Table 2 for summary statistics. Table A1 shows the averages for Occupational Licensing and Shadow Economy for each state.

TABLE 1 Variable names, definitions, and sources.

Variable	Definition	Source
Shadow economy	The size of the state-level shadow economy as a percentage of GDP estimated using a dynamic general equilibrium (DGE) model.	Marshall et al. (2023)
Occupational licensing	Employee-weighted occupational licensing prevalence. The weighted average for licensure prevalence in each state for the coded occupations is calculated by multiplying the license score (0, 0.5, or 1) of each state in each year by the occupation's average proportion of total employment in those occupations.	Sorens et al. (2008)
Occupational licensing (spatial lag)	Spatial lag of Occupational licensing using a contiguity weighting matrix.	Sorens et al. (2008) & authors' calculations
Lower-income licensing	The number of lower-income occupations licensed as a percentage of total lower-income occupations.	Knepper et al. (2022)
Licensing fees	The average fees of lower-income occupational licensing measured in U.S. dollars.	Knepper et al. (2022)
Licensing days	The average estimated calendar days lost to obtain a license.	Knepper et al. (2022)
Licensing exams	The average number of exams needed to obtain a license.	Knepper et al. (2022)
Minimum grade	The average minimum grade needed to obtain a license.	Knepper et al. (2022)
Minimum age	The average minimum age to obtain a license.	Knepper et al. (2022)
Government ideology	State government ideology measured as a weighted average of political ideology and state power over public policy for the governor and each of the two major parties in each legislative chamber.	Berry et al. (1998)
LnGDP	The log of real per capita gross state product (constant 1983–1984 dollars).	UK Center for Poverty Research and authors' calculations
Government size	Total state and local government expenditures as a percentage of GDP.	The Census Bureau and authors' calculations
Education	The percentage of the population 25 years or older that has a bachelor's degree or more.	The Census Bureau
Union coverage	Union coverage measured as the percentage of non-agricultural workers that are covered by a collective bargaining agreement.	Hirsch et al. (2001)
Tax freedom	Tax freedom index measures the state's tax burden and is calculated as an unweighted average of the income and payroll tax index, top marginal income tax rate index, property tax index, and sales tax index. This index is scaled from 0 to 10, where higher values indicate greater freedom from taxation.	Fraser Institute
Race fractionalization	Race fractionalization calculated as $1 - \sum_i s_i^2$ where s_i is the population share of group i where i is American Indian or Alaska Native, Asian or pacific Islander, black or African American, and White.	National Center for Health Statistics (NCHS) and authors' calculations
Democrat governor	Binary variable equal to one if the Governor's political party affiliation is democrat, and zero otherwise.	UK Center for Poverty Research
Fiscal decentralization	Fiscal decentralization measured as the total local own revenue divided by total federal revenue to local governments. This measure of fiscal decentralization is the autonomy indicator from Akai and Sakata (2002).	The Census Bureau and authors' calculations
State minimum wage	State minimum wage (constant 1983–1984 dollars).	UK Center for Poverty Research and authors' calculations

TABLE 2 Summary statistics.

Variable	Mean	Standard deviation	Minimum	Maximum
Shadow economy	0.129	0.055	0.012	0.348
Occupational licensing	0.508	0.126	0.097	0.725
Occupational licensing (spatial lag)	0.471	0.126	0.000	0.687
Lower-income licensing^	0.527	0.148	0.255	0.755
Licensing fees^	281.520	111.118	92	727
Licensing days^	351.980	193.900	113	972
Licensing exams^	1.440	0.497	1	2
Minimum grade^	1.800	0.960	0	4
Minimum age^	9.980	3.285	4	16
LnGDP	9.980	0.185	9.529	10.499
Government size	19.699	3.122	11.920	33.228
Education	28.197	5.231	15.100	45.000
Union coverage	12.284	5.435	2.600	27.900
Tax freedom	5.822	0.933	2.902	8.256
Fiscal decentralization	0.681	0.085	0.374	0.934
Democrat governor	0.433	0.496	0	1
State minimum wage	1.137	0.154	0.255	1.608
Occupational licensing (spatial lag, inverse distance)	0.492	0.054	0.371	0.589
Government ideology	45.441	15.866	17.512	73.619

Note: Summary statistics are based on data for the 50 U.S. states from 2001 to 2019 for a total of 950 observations. The superscript ^ denotes variables that are available only for a cross-section of the 50 states. The variable Government Ideology has only 850 observations.

The primary variable of interest in this study is the size of the state-level shadow economy. Due to the inherent secrecy associated with the shadow economy, measuring its activity is extremely complicated. Individuals working underground take certain precautions not to have their activity recorded due to its illegal nature. Consequently, researchers have attempted to use a variety of statistical tools to come up with reasonable estimates for the size of the shadow economy (see Schneider & Buehn, 2016 for details). Two common model-based methods for estimating the shadow economy are the Multiple Indicators, Multiple Causes (MIMIC) method and the Dynamic General Equilibrium (DGE) method. Both provide a structured framework for analyzing the complexities of the shadow economy, using observable variables and establishing assumptions to guide the models.

One of the first attempts to estimate the size of U.S. state-level shadow economies comes from Wiseman (2013). Wiseman (2013) used a statistical method for dealing with latent variables known as the Multiple Indicators, Multiple Causes (MIMIC) method to estimate the size of state-level shadow economies for the 50 U.S. states from 1997 to 2008. The MIMIC method utilizes a simultaneous equations model comprised of observable variables labeled as either "indicators" or "causal" factors of the shadow economy. The measurement equation relates the latent shadow economy variable to indicator variables and the structural equation relates causal variables to the latent shadow economy. For causal factors Wiseman (2013) uses government size, indirect tax revenues, charges (user fees and fines), labor market freedom, insurance trust expenditures, and regulation activity, while indicators include electricity consumption, growth in labor force participation, and real GDP per capita.[9]

More recently, Marshall et al. (2023) developed a two-sector dynamic deterministic general equilibrium (DGE) model that estimates the size of the shadow economy for all 50 states annually from 1999 to 2019.[10] In particular, this model distinguishes between formal and informal sectors of the economy, with each sector experiencing its own labor-augmenting technological progress based on deterministic trends in time-series variables. This new measure of the shadow economy not only expands the time frame of the previous shadow estimates to 2019 but is also theoretically grounded and includes more time variation in the dynamics of the shadow economy. Of course, similar to the MIMIC model, using DGE modeling to estimate the size of the shadow economy also has its own limitations. For instance, DGE models rely on assumptions that may not be realistic, such as the use of representative agents, perfect foresight, and rational expectations. Furthermore, like the MIMIC method, calibration and parameter selection can also prove to be challenging.[11] Nonetheless, for this paper, we use the measure of the shadow economy from Marshall et al. (2023). Table A2 in the Appendix shows the average size of the shadow economy is 13% of GDP with Georgia (22%), North Carolina (20.3%), Nevada (20.1%), South Carolina (19.5%), and Michigan (19.4%) having the largest shadow economy and New York (7.9%), Alaska (7.4%), Montana (7.1%), North Dakota (5.4%), and Wyoming (4.9%) having the smallest shadow economy.

Figures A1 and A2 in the appendix show the time series of the size of the shadow economy, disaggregated by, respectively, the nine Census divisions and by Blue versus Red states based on the number of electoral votes in the 1996 Presidential elections. Both graphs show that the shadow economy follows a very similar trajectory over time, with a spike during the Great Recession. Interestingly, in Figure A2, the size of the shadow economy began to diverge between blue and red states after the Great Recession, with the shadow economy being larger in blue states from around 2009 until about 2014, then it was smaller than in red states after 2015.

The primary independent variable of interest that is used to explain the shadow economy is Occupational Licensing from Sorens et al. (2008). This variable is measured as the employee-weighted average of each state's licensure incidence. According to Sorens et al. (2008), "A regulation is counted as a 'license' only if it virtually prohibits a natural person from practicing the occupation without first obtaining permission, which in turn depends on either the discretion of a government body or certain training or educational requirements." For each state, each occupation, defined using the six-digit Standard Occupational Classification code, is coded as "0" (not a licensed occupation), "0.5" (states that license contractors but not their employees), or "1" (licensed occupation). Each code each year is then multiplied by the average proportion of total employment in each occupation.[12] Table A2 in the Appendix shows that the weighted average of each state's occupational licensure incidence in the sample is 50%. The states with the highest prevalence of occupational licensing are Tennessee (70%), California (69.6%), Nevada (68.6%), North Dakota (67.1%), and Arkansas (65.8%), whereas Indiana (36.8%), Idaho (35.6%), Colorado (33.2%), Kansas (29.3%), and Missouri (17.4%) have the lowest occupational licensing prevalence. Figure A3 in the appendix documents the positive trajectory over time of occupational licensing prevalence, disaggregated by the nine Census divisions.

The correlation between Shadow Economy and Occupational Licensing shown in Table A1 in the Appendix is positive albeit small (0.11). Likewise, Figure A4 plots the relationship between Occupational Licensing and Shadow economy, which shows a slightly positive association. This cursory look at the data reveals evidence that is consistent with the main hypothesis, which posits a positive association. In the next section, we will take a closer look at this relationship by accounting for other factors that may bias this association and address the possible reverse causality.

The remaining variables, which act as controls, are obtained from reputable sources and are frequently used in existing literature—see, for example, Berdiev et al. (2015) and Goel and Saunoris (2016). For variable details see Tables 1 and 2. To control for state prosperity, we account for the (log of) state-level GDP measured in constant dollars collected from Bureau of Economic Analysis. The variable Government Size controls for the extent to which government, rather than markets, are allocating resources, which is measured as total state and local government expenditures as a percent of GDP. Formal education raises the net return from working in the formal sector relative to the informal sector, therefore, the variable Education is included to control for this aspect. This variable is measured as the percent of the state-level population with at least a bachelor's degree and is collected from the Census Bureau. The last control variable, Union Coverage, accounts for the states with a larger union presence (Hirsch et al., 2001).

We also control for several other relevant factors that may bias the relationship between the shadow economy and occupational licensing. First, Tax Freedom from the Fraser Institute accounts for the extent of tax burden within a state. Second, Race Fractionalization accounts for the degree of racial diversity within a state with the raw race data collected from the National Center for Health Statistics (NCHS). Political aspects are accounted for using a dummy variable for democratic governors, collected from the UK Center for Poverty Research (UKCPR). Third, from the Census Bureau, we include a variable capturing the degree of fiscal decentralization of a state, which is measured as total local government revenue divided by total federal grants to local governments (Akai & Sakata, 2002). Finally, to account for state-level labor regulations we include state minimum wage measured in constant dollars collected from the UKCPR.

3.3 | Estimation

To estimate Equation (1) we use two-stage least squares estimation to account for the endogeneity of Occupational Licensing. For example, policy makers in states with heightened shadow activity where individuals are supplying services underground may be more likely to enact policies like occupational licensing to safeguard consumer health and safety from underground suppliers. To instrument for Occupational Licensing, we use the one-period time lag of the spatial lag of Occupation Licensing. This variable measures the weighted average of neighboring states' level of occupational licensing incidence (for details see Anselin, 1988). In this context, and to maintain exogeneity of the weighting matrix, we define "neighborliness" as contiguous states. The spatial lag of the endogenous variable makes for a good instrument because it is likely highly correlated with the endogenous variable due to evidence that supports strategic interactions among state governments in setting regulatory policies (see, e.g., Fredriksson et al., 2004). Furthermore, we argue that this instrument is exogenous, as changes in occupational licensing in a neighboring state (in a previous year) are unlikely to directly affect own-state's shadow economy.[13] Instead, these changes are more likely to impact own-state's occupational licensing through, for example, policy mimicking, where a state imitates policies in neighboring states (see, e.g., Heyndels & Vuchelen, 1998).[14] To ensure the relevancy of this instrument, we report the first-stage coefficient on the instrument as well as its t-statistic at the bottom of Table 3. Because the first-stage coefficient on the instrument is highly statistically significant and the t-statistics are all greater than the rule of thumb 3.2 (in absolute value), we can conclude that the instrument is highly correlated with the endogenous variable and thus relevant (see Stock & Yogo, 2005).

Additionally, we test for heteroskedasticity, serial correlation, and cross-sectional dependence in the error term and report the test results at the bottom of Table 3. First, the Breusch-Pagan test for heteroskedasticity, under the null of homoskedasticity, shows a rejection of the null in favor of heteroskedasticity across all models. Second, the Breusch-Godfrey test for serial correlation shows a rejection of the null no serial correlation across all models. Finally, the Pesaran's CD test for cross-sectional dependence shows a rejection of cross-sectional independence in favor of cross-sectional dependence across all models. In summary, these tests suggest that the errors are heteroskedastic, serially correlated, and cross-sectionally dependent. Consequently, we correct the standard errors by using Driscoll-Kraay (1998) standard errors. Driscoll-Kraay standard errors are robust to heteroskedasticity, serial correlation, and cross-sectional dependence.

4 | RESULTS

4.1 | Baseline results

The baseline results are reported in Table 3. Consistent with our main hypothesis, the coefficient on Occupational Licensing is positive and statistically significant across all models.[15] In terms of economic significance, in Model 3.1 the elasticity is unit elastic, that is, a 10% increase in occupational licensing increases the size of state-level shadow economy, on average, by 10%, ceteris paribus. These results confirm the adverse consequences of occupational licensing in driving individuals underground to supply their labor or demand goods/services at lower costs. The control variables also mostly match with expectations. The coefficients on LnGDP and Education are negative and highly statistically significant across all models; however, Government Size and Union Coverage lack statistical significance.[16,17]

TABLE 3 The effect of occupational licensing on the shadow economy: Baseline results.

	(3.1)	(3.2)	(3.3)	(3.4)	(3.5)	(3.6)
Occupational licensing	0.259**	0.251**	0.267*	0.265**	0.259**	0.266**
	(0.123)	(0.126)	(0.137)	(0.115)	(0.124)	(0.122)
LnGDP	−0.322***	−0.323***	−0.325***	−0.347***	−0.322***	−0.324***
	(0.041)	(0.042)	(0.040)	(0.040)	(0.042)	(0.040)
Government size	−0.002	−0.002	−0.002	−0.003	−0.002	−0.002
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Education	−0.004***	−0.004***	−0.004***	−0.004***	−0.004***	−0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Union coverage	0.0005	0.0005	0.0004	0.0003	0.0005	0.0004
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Tax freedom		−0.002				
		(0.003)				

TABLE 3 (Continued)

	(3.1)	(3.2)	(3.3)	(3.4)	(3.5)	(3.6)
Race fractionalization			0.268***			
			(0.092)			
Fiscal decentralization				−0.196***		
				(0.039)		
Democrat governor					0.0003	
					(0.001)	
State minimum wage						0.008
						(0.009)
Breusch-Pagan test for heteroskedasticity	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Breusch-Godfrey test for serial correlation	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Pesaran's CD test for cross-sectional dependence	[0.046]	[0.047]	[0.041]	[0.030]	[0.046]	[0.046]
First-stage statistics						
Occupational licensing (spatial lag)	−0.264***	−0.274***	−0.263***	−0.262***	−0.261***	−0.266***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
t statistic	−6.711	−7.131	−6.665	−6.660	−6.507	−6.273
Observations	950	950	950	950	950	950
Adjusted R^2	0.667	0.670	0.667	0.675	0.667	0.665
Overall F statistic	2079***	2096***	2083***	2147***	2076***	2062***

Note: Dependent variable: Shadow Economy. Each model is estimated using the instrumental variables technique for a two-way fixed effects model. State and year effects are accounted for but not reported. The endogenous variable Occupational licensing is instrumented using the one-period lag of its spatial lag (Occupational Licensing (spatial)). Driscoll-Kraay standard errors are in parentheses and probability values are in brackets.

Asterisks denote significance at the following levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.1$.

Models 3.2–3.6 include various other factors that may affect the size of the shadow economy. More racially diverse states, on average, have larger shadow economies, and this result is statistically significant at the 1% level. Conceivably, racial discrimination could shut individuals out of the formal sector and encourage shadow participation. Furthermore, fiscally decentralized states, on average, have smaller shadow economies. Decentralized governments constrain government power and thus reduce the incentive to move underground (Goel & Saunoris, 2016). The coefficients on Tax freedom, Democrat governor, and State minimum wage have all the correct signs but lack statistical significance at conventional levels.

Overall, the results confirm the research hypothesis that occupational licensing is positively associated with the size of the shadow economy. In the following sections, we consider various robustness checks on the baseline findings.[18,19] First, we consider an alternate set of instruments (Section 4.2). Second, we control for state-specific linear time trends (Section 4.3). Third, we account for outlying observations (Section 4.4).

4.2 | Robustness check: Accounting for alternate instruments

Given the inherent challenge in identifying suitable instruments, we consider an alternate set of instruments and re-estimate the baseline models. We consider two alternate instruments, one is the (one-period time lag) spatial lag based on the inverse distance weighting matrix, which weighs states closer in geographic proximity more than distant states. Second is the governments ideology index, which is defined by Berry et al. (1998: 327–328) as "the mean position on the same continuum of the elected public officials in a state, weighted according to the power they have over public policy decisions." To measure the dynamic overall ideology of state "i" in year "t", Berry et al. (1998) create a weighted average of the ideology of the lower chamber (25%), upper chamber (25%), and the governor (50%). Liberal-leaning

TABLE 4 The effect of occupational licensing on the shadow economy: Robustness check: Considering alternate instruments (government ideology and occupational licensing (spatial lag, inverse distance)).

	(4.1)	(4.2)	(4.3)	(4.4)	(4.5)	(4.6)
Occupational licensing	0.203*	0.202*	0.198	0.261**	0.209*	0.206*
	(0.121)	(0.120)	(0.133)	(0.109)	(0.127)	(0.118)
LnGDP	−0.344***	−0.344***	−0.344***	−0.376***	−0.345***	−0.346***
	(0.035)	(0.036)	(0.033)	(0.029)	(0.036)	(0.034)
Government size	−0.003	−0.003	−0.003	−0.004	−0.003	−0.003
	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)
Education	−0.004***	−0.004***	−0.004***	−0.004***	−0.004***	−0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Union coverage	0.0001	0.0001	0.00002	0.00001	0.0001	0.0001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Tax freedom		−0.00000				
		(0.002)				
Race fractionalization			0.265**			
			(0.110)			
Fiscal decentralization				−0.200***		
				(0.058)		
Democrat governor					0.0002	
					(0.001)	
State minimum wage						0.007
						(0.010)
Observations	850	850	850	850	850	850
First-stage F statistic	16.967***	17.121***	17.117***	17.727***	16.535***	17.843***
Adjusted R^2	0.692	0.692	0.695	0.683	0.689	0.691
Overall F statistic	2049***	2048***	2076***	2001***	2033***	2043***

Note: Dependent variable: Shadow Economy. Each model is estimated using the instrumental variables technique for a two-way fixed effects model. State and year effects are accounted for but not reported. The endogenous variable Occupational licensing is instrumented using the one-period lag of its spatial lag (Occupational Licensing (spatial, inverse distance)). Driscoll-Kraay standard errors are in parentheses.

Asterisks denote significance at the following levels: ***$p < 0.01$, **$p < 0.05$, and *$p < 0.1$.

governments with greater power are more likely to successfully enact legislation, including labor regulations like occupational licensing.

Using these two variables as instruments for Occupational Licensing, we re-estimate the baseline models and report the results in Table 4. The first-stage F statistic is highly statistically significant and greater than the rule of thumb 10 in all cases, suggesting that these instruments are relevant. Consistent with the baseline results, the coefficient on Occupational Licensing is positive and statistically significant (at the 10% level) with the one exception being in Model 4.3 the coefficient is marginally insignificant. The control variables and their effect on the shadow economy are all in line with the baseline findings.

4.3 | Robustness check: Accounting for state-specific time trends

To account for unique state-specific trajectories caused by changes not captured by the model, we control for state-specific time trends as an additional robustness check and re-estimate the baseline model. These results are reported in Table 5. Interestingly, the coefficient on Occupational Licensing maintains its sign and statistical significance, but the

TABLE 5 The effect of occupational licensing on the shadow economy: Robustness check: Accounting for state-specific time trends.

	(5.1)	(5.2)	(5.3)	(5.4)	(5.5)	(5.6)
Occupational licensing	0.501*	0.487*	0.516**	0.484*	0.484*	0.497*
	(0.279)	(0.270)	(0.256)	(0.288)	(0.256)	(0.283)
LnGDP	−0.364***	−0.373***	−0.368***	−0.357***	−0.364***	−0.364***
	(0.048)	(0.045)	(0.046)	(0.051)	(0.048)	(0.049)
Government size	−0.002*	−0.002**	−0.002*	−0.002*	−0.002*	−0.002*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Education	−0.002	−0.002	−0.002	−0.002	−0.002	−0.002
	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)	(0.002)
Union coverage	−0.002***	−0.002***	−0.002***	−0.002***	−0.002***	−0.002***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Tax freedom		−0.010***				
		(0.002)				
Race fractionalization			−0.362			
			(0.623)			
Fiscal decentralization				0.055		
				(0.048)		
Democrat governor					−0.002	
					(0.003)	
State minimum wage						0.003
						(0.008)
Observations	950	950	950	950	950	950
Adjusted R^2	0.701	0.708	0.696	0.706	0.706	0.702
Overall F statistic	2524***	2597***	2477***	2577***	2579***	2536***

Note: Dependent variable: Shadow Economy. Each model is estimated using the instrumental variables technique for a two-way fixed effects model and including state-specific time trends. State and year effects and state-specific linear time trends are accounted for but not reported. The endogenous variable Occupational licensing is instrumented using the one-period lag of its spatial lag (Occupational Licensing (spatial)). Driscoll-Kraay standard errors are in parentheses.

Asterisks denote significance at the following levels: ***$p < 0.01$, **$p < 0.05$, and *$p < 0.1$.

magnitude is almost twice the size as in the baseline results reported in Table 3. Furthermore, the control variables show some interesting differences with the baseline results. For instance, the coefficients on Government Size and Union Coverage are both negative and statistically significant, while the coefficients on Education are statistically insignificant. Furthermore, Race Fractionalization and Fiscal Decentralization are no longer statistically significant, while the coefficient on Tax Freedom is negative and statistically significant, which is consistent with expectations (see Cebula, 1997). Despite significant variation in the control variables' influence on the shadow economy compared to the baseline models, the effect of occupational licensing on the shadow economy persists and intensifies after incorporating state-specific time trends, thereby further reinforcing our main hypothesis.

4.4 | Robustness check: Accounting for outliers

Because outliers can distort the relationship between occupational licensing and the shadow economy, as a final robustness check, we reduce the influence of outliers by winsorizing the variables Shadow Economy and Occupational Licensing. Winsorizing generates a new variable by replacing observations smaller (greater) than the top

TABLE 6 The effect of occupational licensing on the shadow economy: Robustness check: Correcting for outliers.

	(6.1)	(6.2)	(6.3)	(6.4)	(6.5)	(6.6)
Occupational Licensing^	0.241**	0.234**	0.248**	0.247**	0.242**	0.248**
	(0.105)	(0.108)	(0.115)	(0.100)	(0.107)	(0.105)
LnGDP	−0.313***	−0.314***	−0.315***	−0.335***	−0.313***	−0.315***
	(0.037)	(0.037)	(0.035)	(0.034)	(0.037)	(0.036)
Government size	−0.002	−0.002	−0.002	−0.003	−0.002	−0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Education	−0.004***	−0.004***	−0.004***	−0.004***	−0.004***	−0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Union coverage	0.0003	0.0003	0.0003	0.0002	0.0003	0.0003
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Tax freedom		−0.002				
		(0.003)				
Race fractionalization			0.228**			
			(0.090)			
Fiscal decentralization				−0.173***		
				(0.045)		
Democrat governor					0.0004	
					(0.001)	
State minimum wage						0.008
						(0.008)
Observations	950	950	950	950	950	950
Adjusted R^2	0.685	0.687	0.684	0.691	0.684	0.682
Overall F statistic	2225***	2242***	2229***	2288***	2222***	2210***

Note: Dependent variable: Shadow Economy^. Each model is estimated using the instrumental variables technique for a two-way fixed effects model. State and year effects are accounted for but not reported. The endogenous variable Occupational licensing is instrumented using the one-period lag of its spatial lag (Occupational Licensing (spatial)). Variables denoted with ^ are winsorized. Driscoll-Kraay standard errors are in parentheses.
Asterisks denote significance at the following levels: ***$p < 0.01$, **$p < 0.05$, and *$p < 0.1$.

(bottom) 1th (99th) percentile with the values of the 1th (99th) percentiles—for details on winsorizing see Barnett and Lewis (1994). By employing winsorized variables, we re-estimate the baseline models and present the results in Table 6. The findings closely align with the baseline results, indicating that outlying observations do not change the main finding.

5 | CONCLUDING REMARKS

Despite the relatively large theoretical and empirical literature showing a positive association between the extent of labor regulations and the size of cross-country shadow economies, there is relatively little research on this relationship done at the state level. Moreover, there is even less research that looks at specific types of labor regulations, like occupational licensing, on state-level shadow economies. Part of the reason for this lack of research is the difficulty in measuring the size of the shadow economy (Samuel et al., 2021). However, recently Marshall et al. (2023) created a new measure of the state-level shadow economy using a two-sector dynamic general equilibrium model, estimating the size of the shadow economy for the 50 U.S. states from 1999 to 2019. Using this unique measure of the state-level shadow economy, we examine the nexus between the extent of occupational licensing within a state and the prevalence of shadow economic activity.

Using instrumental variables and two-stage least squares estimation, we estimate the relationship between the prevalence of occupational licensing and the size of state-level shadow economies for the 50 U.S. states from 2001 to 2019. The results show a positive association between occupational licensing and the size of the shadow economy that is both statistically and economically significant. In terms of the magnitude of the effect, we find that, on average, a 10% increase in occupational licensing prevalence increases the size of the state-level shadow economy by approximately 10%, ceteris paribus. This result is robust after accounting for other factors, alternate instruments, state-specific time trends, and outlying observations.

These results shed light on the possible overlooked secondary consequences of occupational licensing. Although occupational licensing is meant to combat asymmetric information and protect the consumer's health and safety, it comes at the possible cost of encouraging underground activity. The barriers to entry created by occupational licensing increases the return to working underground by bypassing formal occupational licensing regulations/requirements. Moreover, the higher cost associated with goods and services in licensed industries further motivates consumers to shop in underground/unregulated markets. If, for example, occupational licensing is driving individuals underground, then one possible solution would be to create counter-measures that would prevent individuals from moving underground or to help individuals migrate back to formality either through tax cuts, government training, or to re-evaluate occupational licensing to ascertain their net burden, especially for younger, lower-income individuals, or ex-offenders that would be particularly prone to underground work. Of course, over time, the existence of the shadow economy as an alternative to the formal sector may impact legislation in ways that reduce the burden of occupational licensing (Schneider & Enste, 2013, p. 149).

Although we attempted to provide a convincing argument supported by evidence of a positive association between the shadow economy and occupational licensing, this study is not without its limitations. It should be viewed not as the final word but as a starting point for further examination of the relationship between unregulated markets and labor regulations, as well as the secondary consequences associated with licensing. As new, more refined data become available, future research should continue to examine this relationship across different occupations that require a license and consider the degree of burden each license imposes on incentivizing individuals to enter underground markets.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ENDNOTES

1 The 1950's estimate is from Kleiner and Vorotnikov (2018) and the 2022 estimate is from https://www.bls.gov/cps/cpsaat53.htm.

2 https://www.bls.gov/cps/cpsaat53.htm.

3 https://www.archbridgeinstitute.org/state-occupational-licensing-index-2023/.

4 https://reason.com/2023/11/17/kansas-thinks-you-need-1000-hours-of-training-to-remove-hair/.

5 Of 102 lower-income occupations examined by Knepper et al. (2022), the following occupations are licensed in all states including Washington, D.C.: barber, bus driver, cosmetologist, earth driller, emergency medical technician, manicurist, pest control applicator, school bus driver, skin care specialist, truck driver, and vegetation pesticide applicator.

6 Following Schneider and Enste (2013), we define the shadow economy to include all legal market-based activities that are deliberately concealed from the government in order to avoid taxes, labor regulations, and other administrative requirements.

7 Barber and cosmetologist need to be licensed in 50 states, landscape contractor in 47 states, home childcare in 44 states, commercial carpenter/cabinet market in 25 states, painting contractor in 27 states, taxidermist in 28 states, animal breeder in 29 states, animal trainer in 7 states, and tree trimmer in 8 states—see Table 1 in Knepper et al. (2022).

[8] For the Public Choice theory of occupational licensing see Larkin (2016).

[9] We also considered Wiseman's (2013) measure of the shadow economy and re-estimated the baseline models in Table 3. The coefficient on Occupational Licensing is positive, and much larger in magnitude relative to the baseline findings in Table 3, but statistically insignificant. However, the validity of these results is questionable due to labor freedom being used as a causal variable in the MIMIC model to estimate the shadow economy, which obviously overlaps with licensing burdens. Nonetheless, these results are available upon request from the authors.

[10] Their method follows the methodology by Solis-Garcia and Xie (2018). Often, DGE models have been used to understand the interrelationship between formal and informal economies (see, e.g., Busato & Chiarini, 2004).

[11] For a critique of the methods used for estimating the shadow economy, see Thomas (1999).

[12] Employment data are based on national cross-industry estimates collected from the Occupational Employment Statistics from the Bureau of Labor Statistics.

[13] We also performed a balancing test to assess the validity of the instrument by regressing the instrumental variable on several characteristics of the focal state, including demographics, employment, and average earnings. The coefficients were individually statistically insignificant, and the F-test for joint significance was also statistically insignificant. These results confirm the validity of the instrument. We thank an anonymous referee for this suggestion. The results are available upon request.

[14] Others have also employed information from neighboring states/countries as instruments: Eichengreen & Leblang, 2008; De Soysa & Vadlamannati, 2011; Bergh & Nilsson, 2014; Berggren & Nilsson, 2015.

[15] To check if the shock from the Great Recession impacted the results, we re-estimated the baseline results by splitting the sample into periods before and after 2008. The coefficient on Occupational Licensing is positive and statistically significant in both samples and of roughly the same magnitude. Therefore, the main results are robust. We thank an anonymous referee for this suggestion. These results are available upon request from the authors.

[16] Due to the uniqueness of Alaska and Hawaii given they are without contiguous neighboring states, we reran the results in Table 3 using a sample that excludes Alaska and Hawaii. These results, not reported, but available upon request, confirm the baseline findings reported in Table 3.

[17] Conceivably, immigration can also impact the size of the shadow economy, and this omitted variable may distort the relationship between the shadow economy and occupational licensing. Consequently, to check the robustness of the baseline findings in Table 3, we included the log of the number of immigrants measured as a proportion of state-level population. These results are consistent with the baseline findings, and the coefficient on the immigrants variable is statistically insignificant. These results are available upon request. We thank an anonymous referee for this suggestion.

[18] Because the shadow economy is likely mostly influenced by licensing in lower-income occupations, we also consider the percentage of lower-income occupations that require a license, as well as the burdens associated with obtaining licensing. These burdens include average fees, average estimated number of calendar days lost, average number of exams, average minimum grade on exams, and average minimum age, as reported in Table 6 of Knepper et al. (2022). Based on the results reported in Appendix Table A3, we find that the proportion of lower-income occupations requiring a license is positively associated with the size of the shadow economy, consistent with the main findings. In terms of licensing burden, we find that licensing fees, days lost, and minimum age have a positive association with the shadow economy, which is not surprising given they represent the cost (both time and monetary) associated with obtaining a license. Conversely, the number of exams and the minimum grade on exams are negatively associated with the size of the shadow economy. It could be that occupations requiring an exam or minimum grade on exams are under more scrutiny from state and local governments.

[19] We also consider an additional proxy from Teague (2016) for the prevalence and cost of occupational licensing, which is the number of licensing agencies in each state. The pooled OLS estimates using this variable show a positive and statistically significant association between the number of licensing agencies and the size of the shadow economy. These results are consistent with the main findings and are available upon request from the authors.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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APPENDIX A

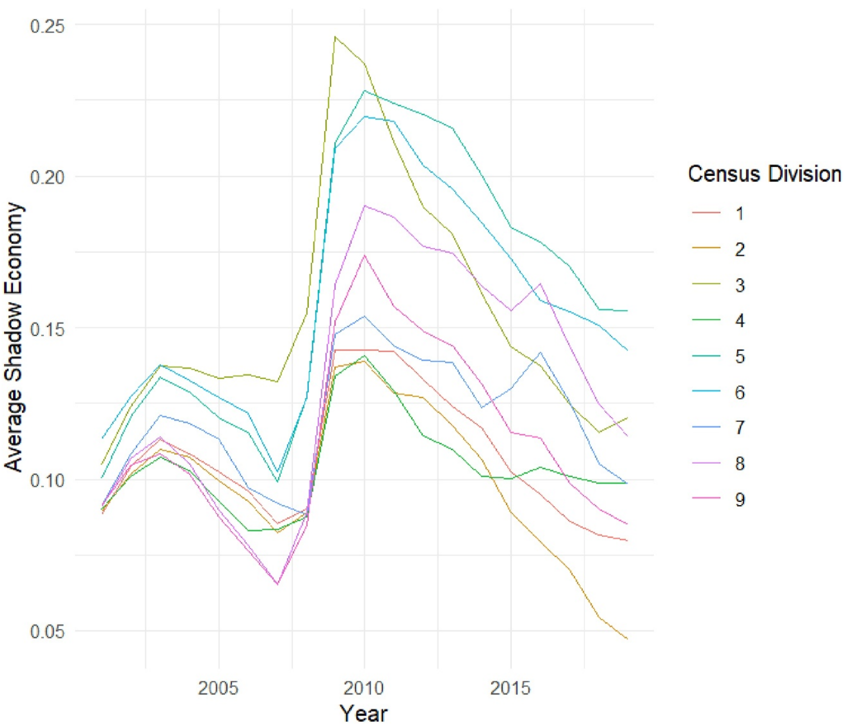

FIGURE A1 The average size of the shadow economy by census division.

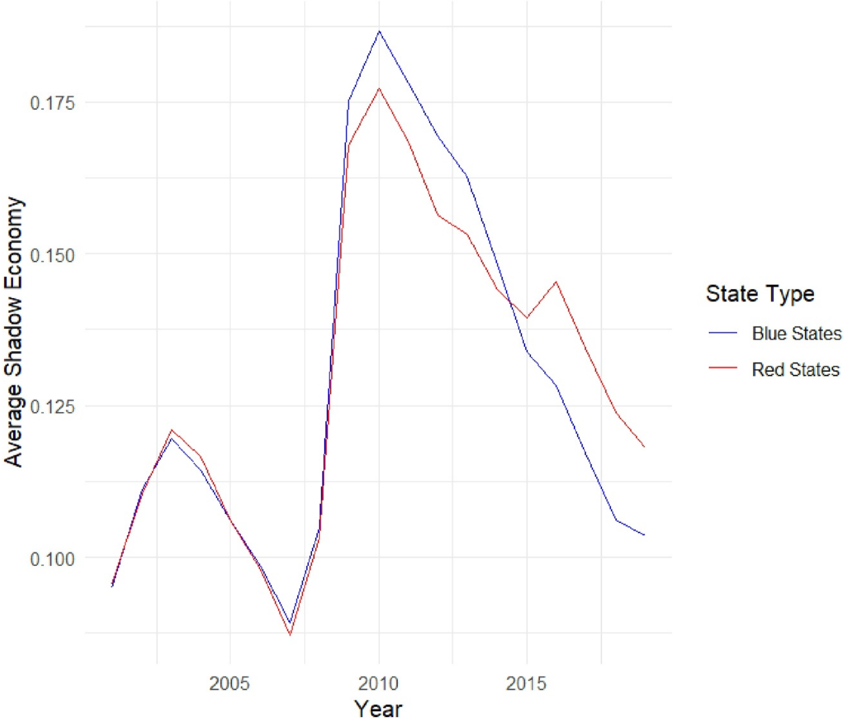

FIGURE A2 The average size of the shadow economy by red versus blue states.

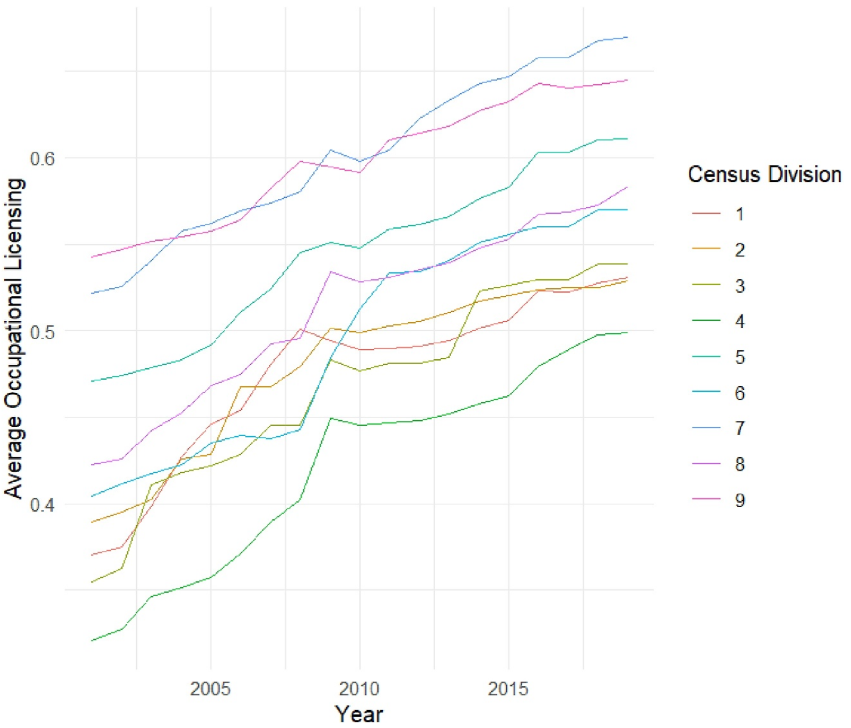

FIGURE A3 The average occupational licensing prevalence by census division.

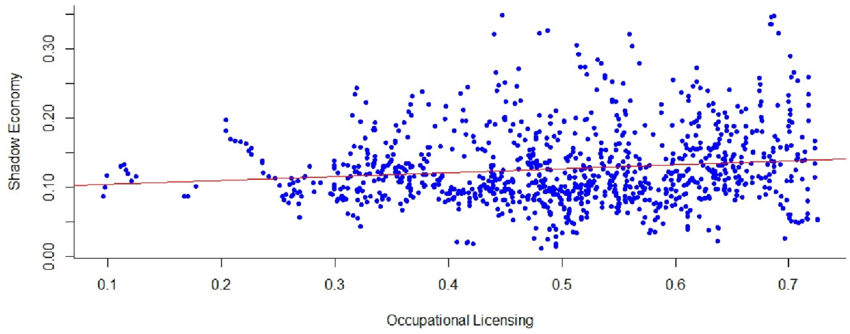

FIGURE A4 Scatterplot.

TABLE A1 Correlation matrix.

	Shadow economy	Occupational licensing	LnGDP	Government size	Education	Union coverage
Shadow economy	1.000					
Occupational licensing	0.112	1.000				
LnGDP	−0.208	0.161	1.000			
Government size	−0.040	0.024	−0.389	1.000		
Education	−0.070	0.127	0.581	−0.309	1.000	
Union coverage	−0.243	−0.071	0.322	0.179	0.260	1.000

Note: $N = 950$.

TABLE A2 Averages of occupational licensing and shadow economy for the 50 U.S. states.

States	Shadow economy (averages)	Rank	Occupational licensing (averages)	Rank
Alabama	0.150	13	0.476	32
Alaska	0.074	47	0.561	18
Arizona	0.181	6	0.623	9
Arkansas	0.140	17	0.658	5
California	0.120	29	0.696	2
Colorado	0.172	9	0.332	48
Connecticut	0.133	25	0.604	12
Delaware	0.174	8	0.428	40
Florida	0.163	10	0.646	6
Georgia	0.223	1	0.538	21
Hawaii	0.090	42	0.515	24
Idaho	0.132	26	0.356	47
Illinois	0.134	23	0.586	15
Indiana	0.161	11	0.368	46
Iowa	0.103	38	0.436	38
Kansas	0.121	28	0.293	49
Kentucky	0.144	15	0.398	43

(Continues)

TABLE A2 (Continued)

States	Shadow economy (averages)	Rank	Occupational licensing (averages)	Rank
Louisiana	0.098	39	0.588	14
Maine	0.112	33	0.430	39
Maryland	0.116	30	0.605	11
Massachusetts	0.115	31	0.477	31
Michigan	0.194	5	0.446	34
Minnesota	0.125	27	0.490	28
Mississippi	0.160	12	0.401	41
Missouri	0.143	16	0.174	50
Montana	0.071	48	0.580	16
Nebraska	0.092	40	0.399	42
Nevada	0.201	3	0.686	3
New Hampshire	0.109	34	0.383	45
New Jersey	0.114	32	0.527	23
New Mexico	0.108	35	0.590	13
New York	0.079	46	0.438	37
North Carolina	0.203	2	0.505	26
North Dakota	0.054	49	0.671	4
Ohio	0.147	14	0.394	44
Oklahoma	0.106	36	0.532	22
Oregon	0.139	18	0.641	7
Pennsylvania	0.103	37	0.474	33
Rhode Island	0.087	44	0.442	35
South Carolina	0.195	4	0.545	19
South Dakota	0.092	41	0.482	30
Tennessee	0.177	7	0.700	1
Texas	0.136	22	0.631	8
Utah	0.137	21	0.491	27
Vermont	0.086	45	0.513	25
Virginia	0.138	19	0.607	10
Washington	0.137	20	0.575	17
West Virginia	0.087	43	0.486	29
Wisconsin	0.134	24	0.543	20
Wyoming	0.049	50	0.442	36
Average	**0.129**		**0.508**	

TABLE A3 The effect of occupational licensing on the shadow economy: Considering lower-income licensing and burdens of licensing.

	(3A.1)	(3A.2)	(3A.3)	(3A.4)	(3A.5)	(3A.6)
Lower-income licensing	0.029*					
	(0.016)					
Licensing fees		0.0001**				
		(0.00004)				
Licensing days			0.00005*			
			(0.00003)			
Licensing exams				−0.007**		
				(0.003)		
Minimum grade					−0.004***	
					(0.001)	
Minimum age						0.002*
						(0.001)
LnGDP	−0.093***	−0.096***	−0.084***	−0.098***	−0.094***	−0.093***
	(0.027)	(0.027)	(0.022)	(0.028)	(0.027)	(0.026)
Government size	−0.005***	−0.005***	−0.004***	−0.005***	−0.005***	−0.005***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Education	−0.0002	−0.0001	−0.0004	−0.0003	−0.0004	−0.0003
	(0.0003)	(0.0003)	(0.0004)	(0.0004)	(0.0003)	(0.0003)
Union coverage	−0.0005	−0.001	−0.001*	−0.0005	−0.0004	−0.001*
	(0.0004)	(0.001)	(0.001)	(0.0003)	(0.0004)	(0.0004)
Observations	950	950	950	950	950	950
Adjusted R^2	0.387	0.410	0.408	0.384	0.386	0.390
Overall F statistic	27.052***	29.694***	29.447***	26.758***	26.974***	27.412***

Note: Dependent variable: Shadow Economy. Each model is estimated using pooled OLS including year dummies that were not reported. Driscoll-Kraay standard errors are in parentheses.

Asterisks denote significance at the following levels: ***$p < 0.01$, **$p < 0.05$, and *$p < 0.1$.